Mohammad Najeeb

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Experience

Educational Technology Department, Saarland University **Apr. 2026 – Present**
Research Assistant (HiWi) – AI & Applied ML *Saarbrücken, Germany*

- **SRL Adaptive Learning Platform (Full-Stack Lead Developer):** Architected and deployed a production learning platform (React 19, Node.js/Express, PostgreSQL 16 with pgvector) serving 50+ university course modules; authored 18 design documents for authentication, LLM provider switching, and data architecture.
- Built document-ingestion pipeline converting lecture PDFs into searchable learning content: PDF parsing to semantic chunking via LLM to vector embeddings (pgvector). Indexed 1200+ documents with HNSW indexing; reduced manual content labeling time by 40%.
- Designed hybrid search combining vector similarity and full-text search (via Reciprocal Rank Fusion), achieving 78% retrieval accuracy on university knowledge base. Deployed as FastAPI service with sub-200ms response latency, enabling real-time student queries.
- Built adaptive AI tutor backend supporting 3 content depth levels per concept, with LLM provider flexibility (OpenAI, Anthropic, Ollama). Implemented cost and latency tracking per model to inform model selection; measured response quality across 500+ student interactions.
- **AI Co-Design Platform (NLP Pipeline):** Built system converting classroom audio to structured knowledge graphs: speech-to-text (Whisper) plus speaker identification (pyannote) feeding into entity and relationship extraction. Processed 45+ hours of lecture audio; outputs versioned in PostgreSQL for team collaboration.
- Implemented flexible LLM provider support (OpenAI, Anthropic Claude, Ollama) with per-call tracking of cost, latency, and output quality; enables comparing model performance and choosing optimal provider based on task requirements.

Computational Interaction Group (CIX), Saarland University **Sept. 2024 – Mar. 2025**
Graduate Research Assistant (HiWi) – ML Pipelines *Saarbrücken, Germany*

- Engineered computer vision ETL pipeline (MediaPipe, OpenCV) processing 50,000+ behavioral events from touchscreen and gesture interaction studies; achieved 96% accuracy on hand-posture classification for real-time event detection, eliminating manual frame annotation.
- Delivered production-ready, reproducible dataset pipeline that reduced manual annotation bottleneck by 30% and enabled PhD-level forecasting research on user interaction patterns.
- Built reusable pipeline architecture enabling scalable behavioral analysis for human-computer interaction (HCI) research across multiple gesture modalities.

Taibah University **Nov. 2021 – May 2022**
Research Intern – Computer Vision & Deep Learning *Medina, Saudi Arabia (Remote)*

- Processed SDNET2018 dataset (56,000+ concrete crack images) for CNN-based detection; improved Matthews Correlation Coefficient (MCC) by 20% using knowledge distillation and interpretability (Grad-CAM) on ResNet-50 and VGG-19. Achieved 96% classification accuracy on test set, enabling more robust automated inspection.

Orion Solutions (ISO 9001:2008 Certified) **Jul. 2022 – Sept. 2023**
Front-End Developer & UI/UX Intern *Gorakhpur, India – Hybrid*

- Shipped responsive, accessible web interfaces (React, HTML, JavaScript, CSS) in cross-functional Agile team; managed design-to-dev handoff with product/UX teams; integrated REST APIs and database layers (MongoDB, SQL).
- Collaborated on end-to-end feature delivery from design mockups through testing, ensuring pixel-perfect implementation and accessibility standards compliance.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
AI/ML: RAG pipelines, LLM integration (OpenAI, Anthropic, Ollama), prompt engineering, vector databases (pgvector), semantic search, fine-tuning, agentic workflows, LangChain, Hugging Face
Backend: FastAPI, Node.js/Express, REST APIs, WebSockets, Celery, Redis
Frontend: React 19, TypeScript, responsive design, accessibility
ML/CV: PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, NumPy, Pandas
Data & DevOps: PostgreSQL, Docker, Git, Linux, CI/CD, NLP (Whisper, speaker diarization, entity extraction)

Education

Universität des Saarlandes **2025 – 2027**
Master of Science in Computer Science (Specialization: Media Informatics) *Saarbrücken, Germany*

Aligarh Muslim University **2019 – 2022**
Bachelor of Science in Mathematics *Aligarh, India*

Publications

- **Zea Mays Leaf Disease Classification Using Swin Transformer** – Springer (SIGMAA 2024) [*DOI: 10.1007/978-981-97-6352-8_59*]

Key Projects

NCBI Gene MCP Client
Production-Ready Bioinformatics API Integration *Python, FastAPI, MCP Protocol*

- Built API server integrating NCBI gene database, deployed on Vercel; enables LLM applications to retrieve gene/protein information in real-time.

UniChatbot
Semantic Search & RAG System *LangChain, ChromaDB, FastAPI, Docker*

- Engineered RAG chatbot serving 500+ university queries; achieved 35% improvement in query relevance via semantic similarity search and retrieval optimization.
- Deployed via Docker; designed for production scalability and easy knowledge base updates.

Languages

- **English:** C1 (Fluent)
- **German:** A2 (Learning)